01 Introduction to LangChain

Original educational material · RAG chunk ID: RAG-001

Core idea

LangChain coordinates language models, prompts, retrievers, and external tools. It reduces integration code but does not replace application design.

Applied example

A PDF question-answering service loads documents, retrieves relevant excerpts, and sends them with a question to a model.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the introduction to langchain stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Compare a direct SDK implementation with a chain that uses interchangeable components.

Retriever test questions

- What does introduction to langchain mean?
- Why is introduction to langchain useful in an AI application?
- What can go wrong when implementing introduction to langchain?

Tags and metadata

chunk_id=RAG-001 | topic=Introduction to LangChain | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

Source inspiration and usage statement

This is original educational material, NOT a YouTube transcript. Topics are inspired by the CampusX LangChain and LLM evaluation series shared in this conversation.

02 LLM Application Architecture

Original educational material · RAG chunk ID: RAG-002

Core idea

Separate ingestion, retrieval, generation, evaluation, and observability. Each stage has its own failures and performance budget.

Applied example

A helpdesk assistant ingests policy files nightly and retrieves excerpts during customer conversations.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the llm application architecture stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Draw component boundaries and specify how failures propagate.

Retriever test questions

- What does llm application architecture mean?
- Why is llm application architecture useful in an AI application?
- What can go wrong when implementing llm application architecture?

Tags and metadata

chunk_id=RAG-002 | topic=LLM Application Architecture | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

03 Embeddings and Semantic Meaning

Original educational material · RAG chunk ID: RAG-003

Core idea

An embedding model maps text to dense numeric vectors. Nearby vectors often indicate related meanings, but closeness is model-dependent.

Applied example

An order-cancellation question can retrieve a document titled steps to void a purchase.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the embeddings and semantic meaning stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Compare semantic similarity against exact keyword search for product identifiers.

Retriever test questions

- What does embeddings and semantic meaning mean?
- Why is embeddings and semantic meaning useful in an AI application?
- What can go wrong when implementing embeddings and semantic meaning?

Tags and metadata

chunk_id=RAG-003 | topic=Embeddings and Semantic Meaning | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

04 Prompts and Instructions

Original educational material · RAG chunk ID: RAG-004

Core idea

Prompt templates combine static instructions with dynamic variables. Clear constraints and explicit context boundaries improve reproducibility.

Applied example

A technical assistant receives a system instruction, retrieved context, and a new user question.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the prompts and instructions stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Test the same prompt with and without retrieved context.

Retriever test questions

- What does prompts and instructions mean?
- Why is prompts and instructions useful in an AI application?
- What can go wrong when implementing prompts and instructions?

Tags and metadata

chunk_id=RAG-004 | topic=Prompts and Instructions | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

05 Chat Messages

Original educational material · RAG chunk ID: RAG-005

Core idea

Role-bearing messages distinguish application instructions, human requests, model outputs, and tool results. History must be managed explicitly.

Applied example

An assistant resolves the word it by examining the latest relevant conversational turns.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the chat messages stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Test a follow-up question with conversation history omitted.

Retriever test questions

- What does chat messages mean?
- Why is chat messages useful in an AI application?
- What can go wrong when implementing chat messages?

Tags and metadata

chunk_id=RAG-005 | topic=Chat Messages | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

06 Structured Output

Original educational material · RAG chunk ID: RAG-006

Core idea

Schema-guided generation returns fields suitable for programmatic consumption. Validate results even when a provider supports constrained output.

Applied example

A support email becomes a JSON object with category, priority, and next action.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the structured output stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Test null fields, unexpected values, and malformed output.

Retriever test questions

- What does structured output mean?
- Why is structured output useful in an AI application?
- What can go wrong when implementing structured output?

Tags and metadata

chunk_id=RAG-006 | topic=Structured Output | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

07 Output Parsers

Original educational material · RAG chunk ID: RAG-007

Core idea

Parsers transform model messages into strings or structured objects. Parsing can fail; use clear error handling and schema validation.

Applied example

A review classifier emits text that a Pydantic parser validates as sentiment and score.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the output parsers stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Compare free-form text with schema-constrained output.

Retriever test questions

- What does output parsers mean?
- Why is output parsers useful in an AI application?
- What can go wrong when implementing output parsers?

Tags and metadata

chunk_id=RAG-007 | topic=Output Parsers | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

08 Chains and Composition

Original educational material · RAG chunk ID: RAG-008

Core idea

A chain connects processing steps so that one output becomes the next input. A simple pipeline may use prompt, model, and parser.

Applied example

A product description is translated, shortened, and converted to a catalog format.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the chains and composition stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Measure added latency for each stage.

Retriever test questions

- What does chains and composition mean?
- Why is chains and composition useful in an AI application?
- What can go wrong when implementing chains and composition?

Tags and metadata

chunk_id=RAG-008 | topic=Chains and Composition | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

09 Runnable Interface

Original educational material · RAG chunk ID: RAG-009

Core idea

The Runnable abstraction standardizes invocation, batches, streams, and asynchronous execution across many LangChain components.

Applied example

The same prompt-model-parser chain supports invoke for one case and batch for an evaluation set.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the runnable interface stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Compare sequential execution with parallel requests under rate limits.

Retriever test questions

- What does runnable interface mean?
- Why is runnable interface useful in an AI application?
- What can go wrong when implementing runnable interface?

Tags and metadata

chunk_id=RAG-009 | topic=Runnable Interface | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

10 Runnable Sequence

Original educational material · RAG chunk ID: RAG-010

Core idea

RunnableSequence executes a dependency-ordered set of operations. Outputs must match the next component's expected input.

Applied example

A search query is rewritten before a retriever fetches documents.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the runnable sequence stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Introduce an incompatible output type and document the error.

Retriever test questions

- What does runnable sequence mean?
- Why is runnable sequence useful in an AI application?
- What can go wrong when implementing runnable sequence?

Tags and metadata

chunk_id=RAG-010 | topic=Runnable Sequence | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

11 Runnable Parallel

Original educational material · RAG chunk ID: RAG-011

Core idea

RunnableParallel runs independent operations from a shared input and combines named results. Parallelism can reduce wall-clock time.

Applied example

One branch summarizes a ticket while another identifies its sentiment.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the runnable parallel stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Check whether concurrent calls exceed provider quotas.

Retriever test questions

- What does runnable parallel mean?
- Why is runnable parallel useful in an AI application?
- What can go wrong when implementing runnable parallel?

Tags and metadata

chunk_id=RAG-011 | topic=Runnable Parallel | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

12 Runnable Branch

Original educational material · RAG chunk ID: RAG-012

Core idea

Conditional routing chooses different workflows based on explicit criteria. Routing can use code or model-generated classifications.

Applied example

Shipping questions use logistics documents while billing questions use invoice documents.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the runnable branch stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Test ambiguous and multi-topic questions.

Retriever test questions

- What does runnable branch mean?
- Why is runnable branch useful in an AI application?
- What can go wrong when implementing runnable branch?

Tags and metadata

chunk_id=RAG-012 | topic=Runnable Branch | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

13 Document Loaders

Original educational material · RAG chunk ID: RAG-013

Core idea

Loaders convert external sources into document objects containing content and metadata. Extraction quality varies by source format.

Applied example

A PDF loader returns page text and page numbers for an internal policy manual.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the document loaders stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Check documents whose pages are scans rather than selectable text.

Retriever test questions

- What does document loaders mean?
- Why is document loaders useful in an AI application?
- What can go wrong when implementing document loaders?

Tags and metadata

chunk_id=RAG-013 | topic=Document Loaders | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

14 Chunking Basics

Original educational material · RAG chunk ID: RAG-014

Core idea

Chunking divides large texts into retrievable pieces. Very small chunks lose context while very large chunks can dilute relevance.

Applied example

A returns policy becomes individual sections about eligibility, exceptions, and timelines.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the chunking basics stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Vary chunk size and measure whether answers remain complete.

Retriever test questions

- What does chunking basics mean?
- Why is chunking basics useful in an AI application?
- What can go wrong when implementing chunking basics?

Tags and metadata

chunk_id=RAG-014 | topic=Chunking Basics | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

15 Chunk Overlap

Original educational material · RAG chunk ID: RAG-015

Core idea

Overlap repeats material at chunk boundaries to retain continuity. Excessive overlap increases storage and duplicate search results.

Applied example

A paragraph spanning two pages appears partially in neighboring chunks.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the chunk overlap stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Compare zero overlap with moderate overlap on boundary questions.

Retriever test questions

- What does chunk overlap mean?
- Why is chunk overlap useful in an AI application?
- What can go wrong when implementing chunk overlap?

Tags and metadata

chunk_id=RAG-015 | topic=Chunk Overlap | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

16 Recursive Splitting

Original educational material · RAG chunk ID: RAG-016

Core idea

Recursive splitting tries separators in an ordered hierarchy to preserve natural text structure where possible.

Applied example

A report is split by paragraphs before sentences and individual characters.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the recursive splitting stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Audit unusually long chunks and separator edge cases.

Retriever test questions

- What does recursive splitting mean?
- Why is recursive splitting useful in an AI application?
- What can go wrong when implementing recursive splitting?

Tags and metadata

chunk_id=RAG-016 | topic=Recursive Splitting | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

17 Metadata Design

Original educational material · RAG chunk ID: RAG-017

Core idea

Metadata such as source, section, date, language, and access level improves filtering, auditing, and citations.

Applied example

An employee policy document has source_uri, effective_date, department, and page.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the metadata design stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Retrieve only current policies permitted for the user.

Retriever test questions

- What does metadata design mean?
- Why is metadata design useful in an AI application?
- What can go wrong when implementing metadata design?

Tags and metadata

chunk_id=RAG-017 | topic=Metadata Design | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

18 Vector Stores

Original educational material · RAG chunk ID: RAG-018

Core idea

A vector store persists embeddings and performs nearest-neighbor search. Stores differ in persistence, filters, indexing, and operational controls.

Applied example

A Chroma prototype retrieves five semantically similar documentation chunks.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the vector stores stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Rebuild the index after changing the embedding model.

Retriever test questions

- What does vector stores mean?
- Why is vector stores useful in an AI application?
- What can go wrong when implementing vector stores?

Tags and metadata

chunk_id=RAG-018 | topic=Vector Stores | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

19 Similarity Functions

Original educational material · RAG chunk ID: RAG-019

Core idea

Cosine similarity measures vector orientation; other indexes use dot product or distance. Scores are not universally comparable.

Applied example

Two phrases can rank differently when vectors are normalized or when another metric is chosen.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the similarity functions stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Report score direction and metric for each store.

Retriever test questions

- What does similarity functions mean?
- Why is similarity functions useful in an AI application?
- What can go wrong when implementing similarity functions?

Tags and metadata

chunk_id=RAG-019 | topic=Similarity Functions | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

20 Vector Index Operations

Original educational material · RAG chunk ID: RAG-020

Core idea

Ingestion requires stable identifiers, insertion, updates, and deletion. Re-indexing is necessary when source data changes.

Applied example

A replaced policy must remove stale chunks before new content becomes searchable.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the vector index operations stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Test duplicate IDs and deleted documents.

Retriever test questions

- What does vector index operations mean?
- Why is vector index operations useful in an AI application?
- What can go wrong when implementing vector index operations?

Tags and metadata

chunk_id=RAG-020 | topic=Vector Index Operations | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

21 Retrievers

Original educational material · RAG chunk ID: RAG-021

Core idea

A retriever maps a query to relevant documents using an underlying search strategy. Its success affects all later generation.

Applied example

A user asks about delivery exceptions, and the retriever returns shipping-policy chunks.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the retrievers stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Measure recall at several values of k.

Retriever test questions

- What does retrievers mean?
- Why is retrievers useful in an AI application?
- What can go wrong when implementing retrievers?

Tags and metadata

chunk_id=RAG-021 | topic=Retrievers | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

22 Top-K Retrieval

Original educational material · RAG chunk ID: RAG-022

Core idea

Top-k controls the number of results returned. More results can improve coverage but may add noise and token costs.

Applied example

A chatbot compares top three versus top eight retrieved chunks for a complicated question.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the top-k retrieval stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Plot answer accuracy against context size.

Retriever test questions

- What does top-k retrieval mean?
- Why is top-k retrieval useful in an AI application?
- What can go wrong when implementing top-k retrieval?

Tags and metadata

chunk_id=RAG-022 | topic=Top-K Retrieval | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

23 Hybrid Search

Original educational material · RAG chunk ID: RAG-023

Core idea

Hybrid search combines lexical matching with semantic retrieval. It is often valuable for IDs, codes, and uncommon vocabulary.

Applied example

A customer asks about SKU XR-205 and a delivery policy in the same question.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the hybrid search stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Compare BM25, vector-only, and hybrid ranking.

Retriever test questions

- What does hybrid search mean?
- Why is hybrid search useful in an AI application?
- What can go wrong when implementing hybrid search?

Tags and metadata

chunk_id=RAG-023 | topic=Hybrid Search | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

24 Maximal Marginal Relevance

Original educational material · RAG chunk ID: RAG-024

Core idea

MMR balances relevance with diversity so the retrieved set is less repetitive. It can improve coverage of multi-aspect queries.

Applied example

A product comparison question retrieves specification, warranty, and delivery sections rather than three similar summaries.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the maximal marginal relevance stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Compare answer coverage with ordinary similarity search.

Retriever test questions

- What does maximal marginal relevance mean?
- Why is maximal marginal relevance useful in an AI application?
- What can go wrong when implementing maximal marginal relevance?

Tags and metadata

chunk_id=RAG-024 | topic=Maximal Marginal Relevance | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

25 Multi-Query Retrieval

Original educational material · RAG chunk ID: RAG-025

Core idea

A query expansion step generates alternate formulations and merges retrieved results. It can help ambiguous phrasing but raises cost.

Applied example

A user asks how to undo a purchase; variants include cancellation, reversal, and refund.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the multi-query retrieval stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Check whether expanded queries introduce unrelated documents.

Retriever test questions

- What does multi-query retrieval mean?
- Why is multi-query retrieval useful in an AI application?
- What can go wrong when implementing multi-query retrieval?

Tags and metadata

chunk_id=RAG-025 | topic=Multi-Query Retrieval | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

26 Contextual Compression

Original educational material · RAG chunk ID: RAG-026

Core idea

Compression removes unrelated passages from retrieved documents before generation. Extraction must preserve necessary qualifiers.

Applied example

A long handbook chapter is reduced to paragraphs about sick leave policy.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the contextual compression stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Check that exceptions and effective dates survive compression.

Retriever test questions

- What does contextual compression mean?
- Why is contextual compression useful in an AI application?
- What can go wrong when implementing contextual compression?

Tags and metadata

chunk_id=RAG-026 | topic=Contextual Compression | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

27 Parent Document Retrieval

Original educational material · RAG chunk ID: RAG-027

Core idea

Small child chunks are searched for precision, while larger parent sections are returned for richer context.

Applied example

A search identifies a single sentence, then includes the whole subsection containing it.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the parent document retrieval stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Compare retrieval precision and generation completeness.

Retriever test questions

- What does parent document retrieval mean?
- Why is parent document retrieval useful in an AI application?
- What can go wrong when implementing parent document retrieval?

Tags and metadata

chunk_id=RAG-027 | topic=Parent Document Retrieval | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

28 RAG Fundamentals

Original educational material · RAG chunk ID: RAG-028

Core idea

Retrieval-augmented generation supplies external information in the model's prompt. It can improve grounding but does not guarantee accuracy.

Applied example

An assistant answers a company-specific holiday question using an approved HR document.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the rag fundamentals stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Ask a question absent from the available corpus and require an abstention.

Retriever test questions

- What does rag fundamentals mean?
- Why is rag fundamentals useful in an AI application?
- What can go wrong when implementing rag fundamentals?

Tags and metadata

chunk_id=RAG-028 | topic=RAG Fundamentals | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

29 RAG Ingestion Pipeline

Original educational material · RAG chunk ID: RAG-029

Core idea

Ingestion loads, cleans, splits, embeds, and indexes information. Quality checks should catch empty pages and repeated headers.

Applied example

Hundreds of operational PDFs are processed as a scheduled batch.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the rag ingestion pipeline stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Record parse errors, chunk counts, and embedding version.

Retriever test questions

- What does rag ingestion pipeline mean?
- Why is rag ingestion pipeline useful in an AI application?
- What can go wrong when implementing rag ingestion pipeline?

Tags and metadata

chunk_id=RAG-029 | topic=RAG Ingestion Pipeline | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

30 RAG Answer Pipeline

Original educational material · RAG chunk ID: RAG-030

Core idea

At runtime, the query is processed, relevant passages are retrieved, and the model generates an answer with citations.

Applied example

A user question retrieves three policy excerpts that support a response.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the rag answer pipeline stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Require each factual statement to map to a cited excerpt.

Retriever test questions

- What does rag answer pipeline mean?
- Why is rag answer pipeline useful in an AI application?
- What can go wrong when implementing rag answer pipeline?

Tags and metadata

chunk_id=RAG-030 | topic=RAG Answer Pipeline | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

31 Tool Definitions

Original educational material · RAG chunk ID: RAG-031

Core idea

A tool describes an executable action through a name, argument schema, and description. The application controls authorization.

Applied example

A read-only inventory lookup checks available product stock.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the tool definitions stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Test missing arguments and unauthorized tool requests.

Retriever test questions

- What does tool definitions mean?
- Why is tool definitions useful in an AI application?
- What can go wrong when implementing tool definitions?

Tags and metadata

chunk_id=RAG-031 | topic=Tool Definitions | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

32 Tool Calling

Original educational material · RAG chunk ID: RAG-032

Core idea

A model can request a tool invocation by returning structured arguments. The calling application must execute and validate it.

Applied example

An assistant requests the current status of a shipment using an approved tracking tool.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the tool calling stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Distinguish tool selection from actual execution.

Retriever test questions

- What does tool calling mean?
- Why is tool calling useful in an AI application?
- What can go wrong when implementing tool calling?

Tags and metadata

chunk_id=RAG-032 | topic=Tool Calling | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

33 Agents and Control

Original educational material · RAG chunk ID: RAG-033

Core idea

Agents decide which tools to call and may iterate through observation and action loops. Guardrails and limits constrain behavior.

Applied example

A service assistant first retrieves policy, then checks a status API, then summarizes findings.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the agents and control stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Set maximum iterations, timeouts, and human approval requirements.

Retriever test questions

- What does agents and control mean?
- Why is agents and control useful in an AI application?
- What can go wrong when implementing agents and control?

Tags and metadata

chunk_id=RAG-033 | topic=Agents and Control | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

34 LLM Evaluations Overview

Original educational material · RAG chunk ID: RAG-034

Core idea

Evaluations measure whether a model or application meets defined quality, safety, and operational goals. Metrics depend on tasks.

Applied example

A question-answering assistant is graded for factuality, relevance, cost, and latency.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the llm evaluations overview stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Define acceptance thresholds before tuning.

Retriever test questions

- What does llm evaluations overview mean?
- Why is llm evaluations overview useful in an AI application?
- What can go wrong when implementing llm evaluations overview?

Tags and metadata

chunk_id=RAG-034 | topic=LLM Evaluations Overview | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

35 Golden Datasets

Original educational material · RAG chunk ID: RAG-035

Core idea

A golden dataset contains curated examples, expected outcomes, and often difficulty labels. Diversity matters more than mere volume.

Applied example

Test cases include simple, ambiguous, conflicting, adversarial, and no-answer questions.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the golden datasets stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Separate tuning examples from held-out evaluation cases.

Retriever test questions

- What does golden datasets mean?
- Why is golden datasets useful in an AI application?
- What can go wrong when implementing golden datasets?

Tags and metadata

chunk_id=RAG-035 | topic=Golden Datasets | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

36 Offline and Online Evals

Original educational material · RAG chunk ID: RAG-036

Core idea

Offline evaluation uses repeatable test cases; online evaluation monitors real interactions. Both are needed to catch different failures.

Applied example

A release passes benchmark questions but production monitoring reveals misunderstood abbreviations.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the offline and online evals stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Turn anonymized production failures into new offline tests.

Retriever test questions

- What does offline and online evals mean?
- Why is offline and online evals useful in an AI application?
- What can go wrong when implementing offline and online evals?

Tags and metadata

chunk_id=RAG-036 | topic=Offline and Online Evals | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

37 Retriever Recall and Precision

Original educational material · RAG chunk ID: RAG-037

Core idea

Recall asks whether needed information was retrieved. Precision asks how much retrieved content is useful, sometimes accounting for ranking.

Applied example

A retriever returns five sections, two useful, and misses a required exception.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the retriever recall and precision stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Evaluate coverage and relevance independently.

Retriever test questions

- What does retriever recall and precision mean?
- Why is retriever recall and precision useful in an AI application?
- What can go wrong when implementing retriever recall and precision?

Tags and metadata

chunk_id=RAG-037 | topic=Retriever Recall and Precision | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

38 Faithfulness and Answer Quality

Original educational material · RAG chunk ID: RAG-038

Core idea

Faithfulness checks whether the answer is supported by given context. Correctness asks whether the answer matches trusted facts.

Applied example

A fluent response invents a cancellation fee absent from the policy.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the faithfulness and answer quality stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Score unsupported assertions separately from incomplete answers.

Retriever test questions

- What does faithfulness and answer quality mean?
- Why is faithfulness and answer quality useful in an AI application?
- What can go wrong when implementing faithfulness and answer quality?

Tags and metadata

chunk_id=RAG-038 | topic=Faithfulness and Answer Quality | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

39 Custom Model Benchmarking

Original educational material · RAG chunk ID: RAG-039

Core idea

Choose models using representative workload tests rather than public leaderboards alone. Compare accuracy, latency, cost, and constraints.

Applied example

Two models generate SQL for identical questions against a safe read-only database.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the custom model benchmarking stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Assess execution results rather than exact SQL string matches.

Retriever test questions

- What does custom model benchmarking mean?
- Why is custom model benchmarking useful in an AI application?
- What can go wrong when implementing custom model benchmarking?

Tags and metadata

chunk_id=RAG-039 | topic=Custom Model Benchmarking | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.

40 RAG Regression and Monitoring

Original educational material · RAG chunk ID: RAG-040

Core idea

Continuous improvement involves tracing, sampled evaluations, failure analysis, and release gates. Document versions and metrics matter.

Applied example

A new embedding model improves average recall but harms rare SKU queries.

How it fits into an AI application

This topic affects how an engineering team designs, tests, or operates retrieval-augmented systems. In a production setting, the team should define inputs, outputs, validation rules, and observable failure signals for the rag regression and monitoring stage. The resulting evidence should be recorded alongside the component version and configuration so experiments remain reproducible.

For retrieval experiments, ask users to express the same information need in several ways: a direct question, a paraphrase, a short keyword search, and a question with unrelated details. A successful retrieval system should return the passage that contains the relevant facts while avoiding unnecessary surrounding material. The correct behavior should be evaluated using explicit evidence rather than the fluency of a generated answer.

Practical test

Block release when critical regression tests fail.

Retriever test questions

- What does rag regression and monitoring mean?
- Why is rag regression and monitoring useful in an AI application?
- What can go wrong when implementing rag regression and monitoring?

Tags and metadata

chunk_id=RAG-040 | topic=RAG Regression and Monitoring | domain=AI engineering | document_type=original tutorial | language=en | suitable_for=semantic-search, keyword-search, citation-tests

Notes for evaluation

When using this page in a benchmark, verify whether the retrieved excerpt directly supports the requested answer. A semantically related passage is not sufficient if it omits a critical condition. Preserve the page identifier in citations and record both successes and near-miss retrievals.